Sequences and Functions 2

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1 Functions

1.1 Power Series

Definition. A **power series** is a series of the form

$$\sum_{n=0}^{\infty} a_n x^n.$$

where $a_0, a_1, \ldots$ are all real numbers, called the coefficients of the power series.

Theorem. For any power series $\sum_{n=0}^{\infty} a_n x^n$ either

1. $\sum_{n=0}^{\infty} |a_n x^n|$ converges $\forall x \in \mathbb{R}$
2. $\sum_{n=0}^{\infty} |a_n x^n|$ diverges $\forall x \in \mathbb{R} \setminus \{0\}$
3. $\exists R \in (0, \infty)$ such that $\sum_{n=0}^{\infty} |a_n x^n|$ converges if $|x| < R$ and diverges if $|x| > R$.

Remark. In Case 3, the sequence may or may not converge if $|x| = R$, and it depends on the sequence $(a_n)_{n \in \mathbb{N}}$.

Definition. (Radius of Convergence)

- If $\sum_{n=0}^{\infty} |a_n x^n|$ converges $\forall x \in \mathbb{R}$, we define the **radius of convergence** to be ∞
- If $\sum_{n=0}^{\infty} |a_n x^n|$ diverges for all $|x| > 0$ we define the radius of convergence to be 0.
- If $\sum_{n=0}^{\infty} |a_n x^n|$ converges $|x| < R$ and diverges $|x| > R$, then the **radius of convergence** is defined as R.

Although we know the radius of convergence exists, we know nothing about it. The next theorem gives us some information about it.

Theorem. For any power series $\sum_{n=0}^{\infty} |a_n x^n|$, if

- $a_n \neq 0 \quad \forall n \in \mathbb{N}_0$ and $R = \lim\limits_{n \to \infty} \frac{|a_n|}{|a_{n+1}|}$ exists, then R is the radius of convergence.

1.2 Exponential and Trigonometic Functions

A power series with radius of convergence $R \in (0, \infty]$ gives a function $f : (-R, R) \to \mathbb{R}$. Many important functions can be defined in this way.

Definition. For $x \in \mathbb{R}$, define

$$\exp(x) := \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

Proposition. The power series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ has infinite radius of convergence, thus the exponential function has domain equal to $\mathbb{R}$.

Proposition.

$$\exp(x+y) = \exp(x)\exp(y).$$

Corollary. For any $x \in \mathbb{R}$,

- $\exp(x) > 0$
- $\exp(-x) = \frac{1}{\exp(x)}$.

Definition. For all $x \in \mathbb{R}$,

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}.$$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}.$$

Remark. We cannot find the radius of convergence using the ratio test, as many terms are equal to 0. However, we can compare these series with the exponential function and show they are bounded by the comparison test.

1.3 Limits of Functions

Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$.

Definition. Let $S \subseteq \mathbb{R}$ and $c \in \mathbb{R}$. We say c is a **cluster point** of S if there exists $(x_n)_{n\in\mathbb{N}}$ in $S \setminus \{c\}$ such that $\lim\limits_{n\to\infty} x_n = c$.

Definition. Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$. Let $L \in \mathbb{R}$ and let c be a cluster point of D.

We say f **converges to** L as x tends to c, or $\lim\limits_{x\to c} f(x) = L$ if

$$\forall \epsilon > 0 \quad \exists \delta > 0 \quad \forall x \in D \quad \text{s.t.}$$

$$\text{if } 0 < |x - c| < \delta \text{ then } |f(x) - L| < \epsilon$$

In the above definition, the definition of $f(c)$ is irrelevent.

Theorem. Let $c \in \mathbb{R}$ be a cluster point of $D \subseteq \mathbb{R}$. Let $f : D \to \mathbb{R}$ and $L \in \mathbb{R}$. Then the following are equivalent:

- $\lim\limits_{x \to c} f(x) = L$
- For any sequence $(x_n)_{n \in \mathbb{N}}$ in $D \setminus \{c\}$

$$\lim_{n \to \infty} x_n = c \Rightarrow \lim_{n \to \infty} f(x_n) = L.$$

Proposition. Let c be a cluster point of $D \subseteq \mathbb{R}$. Suppose there exists a sequence $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ in $D \setminus \{c\}$ such that

- $\lim\limits_{n \to \infty} x_n = c$ and $\lim\limits_{y \to \infty} = c$
- $\lim\limits_{n \to \infty} f(x_n) = A$ and $\lim\limits_{n \to \infty} f(y_n) = B$ with $A \neq B$.

Then we conclude $\lim\limits_{x \to c} f(x)$ cannot exist.

Proposition. Let c be a cluster point of $D \subseteq \mathbb{R}$ and let $f : D \to \mathbb{R}$.

If $\lim\limits_{x \to c} f(x) = L_1$ and $\lim\limits_{x \to c} f(x) = L_2$ then

$$L_1 = L_2.$$

Theorem. (Algebra of limits for functions)

Suppose $D \subseteq \mathbb{R}$ and c is a cluster point of D. Let $f, g : D \to \mathbb{R}$ be two functions.

Suppose $\lim\limits_{x \to c} f(x) = L$ and $\lim\limits_{x \to c} g(x) = M$.

Let $x \in \mathbb{R}$. Then

- $\lim\limits_{x \to c} \ (f(x) + g(x)) = L + M$
- $\lim\limits_{x \to c} \ \alpha f(x) = \alpha L$
- $\lim\limits_{x \to c} \ f(x)g(x) = LM$
- If $M \neq 0$, then $\lim\limits_{x \to c} \frac{f(x)}{g(x)} = \frac{L}{M}$

Theorem. Suppose $D \subseteq \mathbb{R}$, $c \in \mathbb{R}$ is a cluster point of D, $f : D \to \mathbb{R}$ and $\lim\limits_{x \to c} f(x) = L$. Then

- If $f(x) \leq M \quad \forall x \in D$, then $L \leq M$
- If $f(x) \geq M \quad \forall x \in D$, then $L \geq M$.

Theorem. Suppose $D \subseteq \mathbb{R}$, $c \in \mathbb{R}$ is a cluster point of D and $f, g, h : D \to \mathbb{R}$ with

$$f(x) \leq g(x) \leq h(x) \quad \forall x \in D.$$

and

$$\lim_{x \to c} f(x) = L = \lim_{x \to c} h(x).$$

Then $\lim_{x \to c} g(x) = L$.

Now we will consider the limit of a function to ∞.

Definition. Let $D \subseteq \mathbb{R}$, $f : D \to \mathbb{R}$ and $L \in \mathbb{R}$. Suppose there exists a sequence $(a_n)_{n \in \mathbb{N}}$ in D such that $\lim_{n \to \infty} a_n = \infty$.

We say f converges to L as $x \to \infty$ if

$$\forall \epsilon > 0 \quad \exists N \in \mathbb{R} \text{ s.t. } \forall x \in D, \quad x > N$$

$$|f(x) - L| < \epsilon.$$

Hence we write $\lim_{x \to \infty} f(x) = L$.

Definition. Let $D \subseteq \mathbb{R}$, $c \in \mathbb{R}$ be a cluster point of D and $f : D \to \mathbb{R}$

- We say f diverges to ∞ as $x \to c$ if $\forall M \in \mathbb{R}$ $\exists \delta > 0$ such that $\forall x \in D$ with $0 < |x - c| < \delta$ we have $f(x) > M$. In this case, we write $\lim_{x \to c} f(x) = \infty$.
- We say f diverges to $-\infty$ as $x \to c$ if $\forall M \in \mathbb{R}$ $\exists \delta > 0$ such that $\forall x \in D$ with $0 < |x - c| < \delta$ we have $f(x) < M$. In this case, we write $\lim_{x \to c} f(x) = -\infty$.

2 Continuity

Definition. Let $D \subseteq \mathbb{R}$, $c \in D$ and $f : D \to \mathbb{R}$. We say f is continuous at c if

$$\forall \epsilon > 0 \quad \exists \delta > 0 \quad \forall x \in D, \quad |x - c| < \delta$$

$$|f(x) - f(c)| < \epsilon.$$

We say f is **continuous** on D if this holds true at every $c \in D$.

Theorem. Let $D \subseteq \mathbb{R}$, $f : D \to \mathbb{R}$ and c be a cluster point of D.

Then f is continuous at c if and only if

$$\lim_{x \to c} f(x) = f(c).$$

Remark. The relation between continuity at c and $\lim_{x \to c} f(x) = f(c)$ does not hold if c is not a cluster point. However, functions are always continuous at points that are not cluster points of the domain.

Theorem. (Algebra of continuous functions)
Let $D \subseteq \mathbb{R}$, $c \in D$ and $\alpha \in \mathbb{R}$. Let $f, g : D \to \mathbb{R}$ be continuous at c, then

- $f + g$
- αf
- fg

are all continuous at c.
If $g(c) \neq 0$, then $\frac{f}{g}$ is also continuous at c.

2.1 Composition of Continuous Functions

Given two sets $D, E \subseteq \mathbb{R}$, $f : D \to \mathbb{R}$ and $g : E \to D$, we can define the composite function
$$(f \circ g)(x) := f(g(x)).$$

Theorem. (Continuity of Continuous Functions)
Let $D, E \subseteq \mathbb{R}$, and $c \in E$. Let $f : D \to \mathbb{R}$ and $g : E \to D$.
If g is continuous at c and f is continuous at $g(c)$, then $f \circ g$ is continuous at c.

2.2 Continuity of Selected Functions

Proposition. The following functions are all continuous.

- $f(x) = \alpha$ for $\alpha \in \mathbb{R}$.
- $f(x) = x$
- $f(x) = \sum_{i=n}^{\infty} a_n x^n$
- $f(x) = \exp(x)$
- $f(x) = \sinh(x)$
- $f(x) = \cosh(x)$
- $f(x) = \sin(x)$
- $f(x) = \cos(x)$
- $f(x) = \sec(x), g(x) = \tan(x)$ where $f, g : \mathbb{R} \setminus \{\frac{\pi}{2} + n\pi | n \in \mathbb{Z}\} \to \mathbb{R}$

- $f(x) = \csc(x), g(x) = \cot(x)$ where $f, g : \mathbb{R} \setminus \{n\pi | n \in \mathbb{Z}\} \to \mathbb{R}$

2.3 The Intermediatve Value Theorem

Let $D \subseteq \mathbb{R}$, $\alpha \in \mathbb{R}$ and $f : D \to \mathbb{R}$. Solving the equation $f(x) = \alpha$ can rarely be done explicitly. However, if f is continuous on $D = [a, b]$, and if α is between $f(a)$ and $f(b)$, it is plausible that $f(x) = \alpha$ has a solution in D.

Theorem. Let $a, b \in \mathbb{R}$ with $a < b$. Suppose

- $f : [a, b] \to \mathbb{R}$ is continuous
- $f(a) \leq \alpha \leq f(b)$ or $f(b) \leq \alpha \leq f(a)$.

Then there exists $c \in [a, b]$ such that

$$f(c) = \alpha.$$

2.4 The Weierstrass Extreme Value Theorem

Definition. Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$.

- We say $c \in D$ is a **minimum point** of f in D if

$$f(c) \leq f(x) \quad \forall x \in D.$$

- We say $c \in D$ is a **maximum point** of f in D if

$$f(c) \geq f(x) \quad \forall x \in D.$$

We say f **attains** its maximum or minimum in D if a maximum or minimum point exists in D.

Theorem. Let $a, b \in \mathbb{R}$ with $a < b$. Suppose $f : [a, b] \to \mathbb{R}$ is continuous. Then

- there exists finite $L > 0$ such that

$$|f(x)| \leq L \quad \forall x \in [a, b].$$

- f attains its maximum and minimum in $[a, b]$.

2.5 The Natural Logarithm

Before we can define the inverse of the exponential function, we must show that it is a bijection.

Proposition. The exponential function is bijective.

Definition. The function $\log : (0, \infty) \to \mathbb{R}$ is the inverse of $\exp : \mathbb{R} \to (0, \infty)$.

Proposition. For $x, y \in (0, \infty)$

- $\log(xy) = \log x + \log y$
- $\log(\frac{x}{y}) = \log x - \log y$

Definition. For $a > 0$ and $x \in \mathbb{R}$

$$a^x = \exp(x \log a).$$

3 Derivatives

Definition. Suppose $S \subseteq \mathbb{R}$. A point $c \in S$ is called an **interior point** of S if there exists $\delta > 0$ such that

$$(c - \delta, c + \delta) \subseteq S.$$

Definition. Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$. Suppose $c \in D$ is an interior point of D. We say f is **differentiable** at c if

$$\lim_{h \to 0} \frac{f(c+h) - f(c)}{h} \quad \text{exists.}$$

The value $f'(c) := \lim_{h \to 0} \frac{f(c+h)-f(c)}{h}$ is called **the derivative** of f at c.

If f is differentiable at every point $c \in D$, we say f is **differentiable** on D.

Proposition. Let $D \subseteq \mathbb{R}$ and let $c \in D$ be an interir point of D. If $f : D \to \mathbb{R}$ is differentiable at c, then it is continuous at c.

3.1 The Algebra of Derivatives

Theorem. Let $D \subseteq \mathbb{R}$ and let $c \in D$ be an interior point. Suppose that $f, g : D \to \mathbb{R}$ are differentiable at c. Then

- $f + g$ is differentiable at c with

$$(f + g)'(c) = f'(c) + g'(c).$$

- $f - g$ is differentiable at c with

$$(f - g)'(c) = f'(c) - g'(c).$$

- αf is differentiable at c with $(\alpha f)'(c) = \alpha f'(c)$
- (Product rule) fg is differentiable at c with

$$(fg)'(c) = f'(c)g(c) - f(c)g'(c).$$

- (Quotient rule). If $g(c) \neq 0$ then $\frac{f}{g}$ is differentiable at c with

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

3.2 Derivatives of Basic Functions

Proposition. Let $f : D \to \mathbb{R}, f(x) = x^2$ where $D = \mathbb{R}$ if $n > 0$ and $\mathbb{R} \setminus \{0\}$ otherwise. Then f is differentiable on D with derivative

$$f'(x) = nx^{n-1}.$$

Corollary. Let $p : \mathbb{R} \to \mathbb{R}$ be a polynomial with

$$P(x) = a_n x^n + ... + a_1 x + a_0.$$

Then P is differentiable on $\mathbb{R}$ and

$$p'(x) = na_n x^{n-1} + ... + a_1.$$

Proposition. The functions $\exp, \cos, \sin$ are differentiable with

- $\exp'(x) = \exp(x)$
- $\cos'(x) = -\sin(x)$
- $\sin'(x) = \cos(x)$

3.3 The Chain Rule

Theorem. Let $D, E \subseteq \mathbb{R}$, $f : D \to \mathbb{R}$ and $g : E \to D$. Let $c \in E$ be an interior point, such that $g(c)$ is an interior point of D. If

- g is differentiable at c
- f is differentiable at $g(c)$
-

then $f \circ g$ is differentiable at c with

$$(f \circ g)'(c) = f'(g(c))g'(c).$$

3.4 The derivative of inverse functions

Theorem. Let $D, E \subseteq \mathbb{R}$ and let $f : D \to E$ bijective function. Suppose $c \in D$ and $f(c) \in E$ are interior points. If f is differentiable at c with $f'(c) \neq 0$ then f^{-1} is differentiable at $f(c)$ with

$$(f^{-1})'(f(c)) = \frac{1}{f'(c)}.$$

4 Working With Derivatives

4.1 Extremal Points

Definition.

- $c \in D$ is called a **local maximum point** of $f : D \to \mathbb{R}$ if there exists $\delta > 0$ such that for all $x \in D$ with $|x - c| < \delta$ we have

$$f(x) \leq f(c).$$

- $c \in D$ is called a **local minimum point** of $f : D \to \mathbb{R}$ if there exists a $\delta > 0$ such that for all $x \in D$ with $|x - c| < \delta$ we have

$$f(c) \leq f(x).$$

Theorem. Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$. Suppose

- c is an interior point
- c is either a local maximum or a local minimum
- f is differentiable at c.

Then

$$f'(c) = 0.$$

4.2 The Mean Value Theorem

Theorem. (Rolle's Theorem)
Let $a < b$ and $f : [a, b] \to \mathbb{R}$. Suppose f is continuous on $[a, b]$ and differentiable on (a, b). If $f(a) = f(b)$ then there exists $c \in (a, b)$ with $f'(c) = 0$.

Theorem. (Mean Value Theorem)
Let $a < b$ and $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Corollary. Let $a < b$ and $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). If $f'(x) = 0$ for all $x \in (a, b)$ then f is constant on $[a, b]$.

Theorem. Let $a < b$, and $f, g : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that

$$(f(b) - f(a))(g'(c)) = (g(b) - g(a))f'(c)$$

or alternatively that

$$\frac{f'(c)}{g'(c)} = \frac{f(c) - f(a)}{g(b) - g(a)}.$$

4.3 Monotonicity

Definition. Let $D \subseteq \mathbb{R}$ be an interval and $f : D \to \mathbb{R}$.

- f is **increasing** on D if

$$x < y \Rightarrow f(x) \leq f(y).$$

- f is **decreasing** D if

$$x < y \Rightarrow f(x) \geq f(y).$$

- f is strictly **increasing** on D if

$$x < y \Rightarrow f(x) < f(y).$$

- f is **decreasing** D if

$$x < y \Rightarrow f(x) < f(y).$$

If any of these conditions are satisfied we say f is **monotone**.

Remark. If $f : D \to \mathbb{R}$ is strictly increasing or decreasing, then it follows that $f(x) \neq f(y)$ if $x \neq y$. Hence this function is injectvie.

Theorem. Let $a < b$ and $f : [a, b] \to \mathbb{R}$. Suppose f is continuous on $[a, b]$ and differentiable on (a, b).

- f is increasing in $[a, b]$ if and only if $f'(x) \geq 0$ for all $x \in (a, b)$.
- f is decreasing in $[a, b]$ if and only if $f'(x) \leq 0$ for all $x \in (a, b)$
- f is strictly increasing in $[a, b]$ if $f'(x) > 0$ for all $x \in (a, b)$

- f is strictly decreasing in $[a,b]$ if $f'(x) < 0$ for all $x \in (a,b)$

Corollary. Let c be an interior point of $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$ differentiable on $(c-r) \cup (c, c+r)$ for $r > 0$.

- If $f'(x) \geq 0$ for $x \in (c-r)$ and $f'(x) \leq 0$ for $x \in (c+r)$, c is a local maximum of f.
- If $f'(x) \leq 0$ for $x \in (c-r)$ and $f'(x) \geq 0$ for $x \in (c+r)$, c is a local minimum of f.

4.4 L'Hôpital's Rule

Theorem. Suppose $f, g : (a,b) \setminus \{c\} \to \mathbb{R}$ are differentiable and satisfy

$$\lim_{x \to c} f(x) = 0 \quad \text{and} \quad \lim_{x \to c} g(x) = 0$$

but $g(x) \neq 0$ and $g'(x) \neq 0$ for all $x \in (a,b) \setminus \{c\}$.
If $\lim\limits_{x \to c} \frac{f'(x)}{g'(x)}$ is defined then

$$\lim_{x \to c} \frac{f(x)}{g(x)} = \lim_{x \to c} \frac{f'(x)}{g'(x)}.$$